

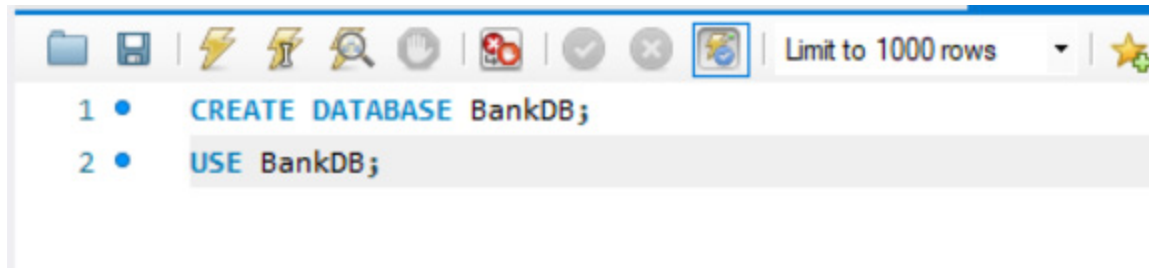
## Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030414

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Query 1: -CREATE DATABASE



Output: -

| #   | Time     | Action                 | Message           | Duration / Fetch |
|-----|----------|------------------------|-------------------|------------------|
| ✓ 1 | 14:24:44 | CREATE DATABASE BankDB | 1 row(s) affected | 0.031 sec        |
| ✓ 2 | 14:24:47 | USE BankDB             | 0 row(s) affected | 0.000 sec        |

Query 2: - CREATE CUSTOMER TABLE

```
• CREATE TABLE Customers (  
    Customer_ID INT PRIMARY KEY,  
    Customer_Name VARCHAR(100) NOT NULL,  
    Phone VARCHAR(15),  
    Email VARCHAR(100),  
    City VARCHAR(50)  
);
```

Output: -

|      |          |                                                                                           |                   |
|------|----------|-------------------------------------------------------------------------------------------|-------------------|
| ✓ 14 | 14:37:28 | CREATE TABLE Customers ( Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT N... | 0 row(s) affected |
|------|----------|-------------------------------------------------------------------------------------------|-------------------|

Query 3: - CREATE ACCOUNT TABLE

```

);
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2) DEFAULT 0,
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);

```

Output: -

|   |   |          |                                              |                   |           |
|---|---|----------|----------------------------------------------|-------------------|-----------|
| ✓ | 4 | 14:28:07 | CREATE TABLE Account ( Account_No INT PRI... | 0 row(s) affected | 0.063 sec |
|---|---|----------|----------------------------------------------|-------------------|-----------|

Query 4: - CREATE TRANSACTION TABLE

```

);
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);

```

Output: -

|   |   |          |                                                |                   |           |
|---|---|----------|------------------------------------------------|-------------------|-----------|
| ✓ | 5 | 14:29:36 | CREATE TABLE Bank_Transaction ( Transaction... | 0 row(s) affected | 0.078 sec |
|---|---|----------|------------------------------------------------|-------------------|-----------|

Query 5: - CREATE LOAN TABLE

```

0 ● ○ CREATE TABLE Loan (
1     Loan_ID INT PRIMARY KEY,
2     Customer_ID INT,
3     Loan_Type VARCHAR(30),
4     Loan_Amount DECIMAL(12,2),
5     Interest_Rate DECIMAL(5,2),
6
7     FOREIGN KEY (Customer_ID)
8     REFERENCES Customer(Customer_ID)
9 );

```

Output: -

|   |   |          |                                             |                   |           |
|---|---|----------|---------------------------------------------|-------------------|-----------|
| ✓ | 6 | 14:30:49 | CREATE TABLE Loan ( Loan_ID INT PRIMARY ... | 0 row(s) affected | 0.031 sec |
|---|---|----------|---------------------------------------------|-------------------|-----------|

Query 6: - INSERT CUSTOMER DATA

```

INSERT INTO customers (customerId, customerName, phone, email, city) VALUES
(101, 'Yishaani', '1234535235', 'yishaani@gmail.com', 'Hyderabad'),
(102, 'Rahul', '9876543210', 'rahul@gmail.com', 'Mumbai'),
(103, 'Priya', '9123456789', 'priya@gmail.com', 'Delhi'),
(104, 'Arjun', '9988776655', 'arjun@gmail.com', 'Bangalore'),
(105, 'Ananya', '8765432109', 'ananya@gmail.com', 'Chennai'),
(106, 'Rohan', '7894561230', 'rohan@gmail.com', 'Pune'),
(107, 'Sai', '9345678120', 'sai@gmail.com', 'Hyderabad'),
(108, 'Neha', '8899776655', 'neha@gmail.com', 'Kolkata'),
(109, 'Vikram', '7654321098', 'vikram@gmail.com', 'Jaipur'),
(110, 'Kavya', '9012345678', 'kavya@gmail.com', 'Vijayawada');

```

Output: -

|   |   |          |                                                                                                            |                   |                                      |
|---|---|----------|------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------|
| ✓ | 7 | 14:33:18 | INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Aswitha Patel', '98... | 5 row(s) affected | Records: 5 Duplicates: 0 Warnings: 0 |
|---|---|----------|------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------|

Query 7: - INSERT ACCOUNT DATA

```

INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad');

```

Output: -

```
16 14:40:14 INSERT INTO Account (Account_No, Customer_ID, ... 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 0.031 sec
```

Query 8:- INSERT TRANSACTION DATA

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000);
```

Output: -

```
17 14:41:48 INSERT INTO Bank_Transaction (Account_No, Tr... 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 0.031 sec
```

Query 9: - INSERT LOAN DATA

```
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),
(502, 102, 'Education Loan', 1000000, 6.5),
(503, 103, 'Car Loan', 800000, 8.2),
(504, 104, 'Personal Loan', 500000, 10.5);
```

Output: -

```
18 14:43:11 INSERT INTO Loan (Loan_ID, Customer_ID, Loan... 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 0.047 sec
```

Query 10: - DISPLAY TABLE DATA

```
SELECT * FROM Customers;
```

Output: -

|   |      |                  |            |                   |           |
|---|------|------------------|------------|-------------------|-----------|
|   | 103  | Akshara Reddy    | 9876543212 | akshara@gmail.com | Bangalore |
|   | 104  | Anirudh Reddy    | 9876543213 | anirudh@gmail.com | Chennai   |
|   | 105  | Ananya Varshitha | 9876543214 | ananya@gmail.com  | Hyderabad |
| * | NULL | NULL             | NULL       | NULL              | NULL      |

```
SELECT * FROM Account;
```

|   | Account_No | Customer_ID | Account_Type | Balance   | Branch     |
|---|------------|-------------|--------------|-----------|------------|
| ▶ | 10001      | 101         | Savings      | 50000.00  | Hyderabad  |
|   | 10002      | 102         | Savings      | 75000.00  | Vijayawada |
|   | 10003      | 103         | Current      | 120000.00 | Bangalore  |
|   | 10004      | 104         | Savings      | 45000.00  | Chennai    |
|   | 10005      | 105         | Current      | 90000.00  | Hyderabad  |
| • | NULL       | NULL        | NULL         | NULL      | NULL       |

```
SELECT * FROM Bank_Transaction;
```

|   | Transaction_ID | Account_No | Transaction_Type | Amount   | Transaction_Date    |
|---|----------------|------------|------------------|----------|---------------------|
| ▶ | 1              | 10001      | DEPOSIT          | 10000.00 | 2026-08-15 14:41:48 |
|   | 2              | 10002      | DEPOSIT          | 15000.00 | 2026-08-15 14:41:48 |
|   | 3              | 10003      | WITHDRAW         | 20000.00 | 2026-08-15 14:41:48 |
|   | 4              | 10004      | DEPOSIT          | 5000.00  | 2026-08-15 14:41:48 |
|   | 5              | 10005      | WITHDRAW         | 10000.00 | 2026-08-15 14:41:48 |
| • | NULL           | NULL       | NULL             | NULL     | NULL                |

```
SELECT * FROM Loan;
```

|   | Loan_ID | Customer_ID | Loan_Type      | Loan_Amount | Interest_Rate |
|---|---------|-------------|----------------|-------------|---------------|
| ▶ | 501     | 101         | Home Loan      | 5000000.00  | 7.50          |
|   | 502     | 102         | Education Loan | 1000000.00  | 6.50          |
|   | 503     | 103         | Car Loan       | 800000.00   | 8.20          |
|   | 504     | 104         | Personal Loan  | 500000.00   | 10.50         |
| • | NULL    | NULL        | NULL           | NULL        | NULL          |

Query 11: - PROCEDURE TO FIND CUSTOMER ACCOUNTS

```

DELIMITER //
CREATE PROCEDURE GetCustomerAccounts(
    IN p_Customer_ID INT
)
BEGIN
    SELECT
        C.Customer_ID,
        C.Customer_Name,
        A.Account_No,
        A.Account_Type,
        A.Balance,
        A.Balance,
        A.Branch
    FROM Customer C
    JOIN Account A
        ON C.Customer_ID = A.Customer_ID
    WHERE C.Customer_ID = p_Customer_ID;
END //
DELIMITER ;
CALL GetCustomerAccounts(101);

```

Output: -

```

✓ 24 14:52:47 CREATE PROCEDURE GetCustomerAccounts( IN p_Customer_ID INT ) BEGIN SELECT C.Cust... 0 row(s) affected
✓ 25 14:52:51 CALL GetCustomerAccounts(101) 1 row(s) returned

```

| Account_No | Account_Type | Balance  | Branch    |
|------------|--------------|----------|-----------|
| 10001      | Savings      | 50000.00 | Hyderabad |

Query 12: - PROCEDURE TO DEPOSIT MONEY

```
DELIMITER //
```

```
CREATE PROCEDURE DepositMoney(  
    IN p_Account_No INT,  
    IN p_Amount DECIMAL(12,2)  
)  
  
BEGIN  
  
    UPDATE Account  
    SET Balance = Balance + p_Amount  
    WHERE Account_No = p_Account_No;  
  
END //  
DELIMITER ;  
CALL DepositMoney(10001, 5000);
```

Output: -

|   |    |          |                                          |                   |
|---|----|----------|------------------------------------------|-------------------|
| ✓ | 25 | 14:55:51 | CALL GetCustomerAccounts(10001)          | 1 row(s) returned |
| ✓ | 26 | 14:55:53 | CREATE PROCEDURE DepositMoney( IN p_A... | 0 row(s) affected |
| ✓ | 27 | 14:56:10 | CALL DepositMoney(10001, 5000)           | 1 row(s) affected |

Query 13: - PROCEDURE TO WITHDRAW MONEY

```
DELIMITER //
```

```
CREATE PROCEDURE WithdrawMoney(  
    IN p_Account_No INT,  
    IN p_Amount DECIMAL(12,2)  
)  
  
BEGIN  
  
    UPDATE Account  
    SET Balance = Balance - p_Amount  
    WHERE Account_No = p_Account_No;  
  
END //
```

```
DELIMITER ;
```

```
CALL WithdrawMoney(10001, 3000);
```

Output: -

|   |    |          |                                         |                   |
|---|----|----------|-----------------------------------------|-------------------|
| ✓ | 28 | 14:57:56 | CREATE PROCEDURE WithdrawMoney( IN p... | 0 row(s) affected |
| ✓ | 29 | 14:58:00 | CALL WithdrawMoney(10001, 3000)         | 1 row(s) affected |

Query 14: - TRIGGER – PREVENT NEGATIVE DEPOSIT

```
DELIMITER //
```

```
• CREATE TRIGGER CheckTransactionAmount
  BEFORE INSERT ON Bank_Transaction
  FOR EACH ROW
  BEGIN
    IF NEW.Amount <= 0 THEN
      SIGNAL SQLSTATE '45000'
      SET MESSAGE_TEXT =
        'Transaction amount must be greater than zero';
    END IF;
  END //
```

```
DELIMITER ;
```

Output: -

|   |    |          |                                            |                   |
|---|----|----------|--------------------------------------------|-------------------|
| ✓ | 30 | 14:59:57 | CREATE TRIGGER CheckTransactionAmount B... | 0 row(s) affected |
|---|----|----------|--------------------------------------------|-------------------|

Query 15: - CREATE TRANSACTION AUDIT TABLE

```
CREATE TABLE Transaction_Audit (
  Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
  Transaction_ID INT,
  Account_No INT,
  Transaction_Type VARCHAR(20),
  Amount DECIMAL(12,2),
  Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

Output: -

|   |    |          |                                               |                   |           |
|---|----|----------|-----------------------------------------------|-------------------|-----------|
| ✓ | 31 | 15:01:02 | CREATE TABLE Transaction_Audit ( Audit_ID ... | 0 row(s) affected | 0.062 sec |
|---|----|----------|-----------------------------------------------|-------------------|-----------|

Query 16: - TRIGGER – TRANSACTION AUDIT

```
DELIMITER //
```

- ```
CREATE TRIGGER TransactionAudit
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
    INSERT INTO Transaction_Audit
    (
        Transaction_ID,
        Account_No,
        Transaction_Type,
        Amount
    )
    VALUES
    (
        NEW.Transaction_ID,
        NEW.Account_No,
        NEW.Transaction_Type,
        NEW.Amount
    );
END //
```

```
DELIMITER ;
```

Output: -

|   |             |                                               |                   |           |
|---|-------------|-----------------------------------------------|-------------------|-----------|
| ✓ | 31 15:01:02 | CREATE TABLE Transaction_Audit ( Audit_ID ... | 0 row(s) affected | 0.062 sec |
|---|-------------|-----------------------------------------------|-------------------|-----------|

Test :-

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 2500);
```

Check: -

|   |             |                                               |                   |           |
|---|-------------|-----------------------------------------------|-------------------|-----------|
| ✓ | 32 15:04:10 | INSERT INTO Bank_Transaction (Account_No, ... | 1 row(s) affected | 0.031 sec |
|---|-------------|-----------------------------------------------|-------------------|-----------|

Query 17: - PROCEDURE – ACCOUNT TRANSFER

DELIMITER //

CREATE PROCEDURE TransferMoney(  
 IN SenderAccount INT,  
 IN ReceiverAccount INT,  
 IN TransferAmount DECIMAL(12,2)  
)

BEGIN

DECLARE SenderBalance DECIMAL(12,2);

SELECT Balance  
 INTO SenderBalance  
 FROM Account  
 WHERE Account\_No = SenderAccount;

IF SenderBalance < TransferAmount THEN

SIGNAL SQLSTATE '45000'  
 SET MESSAGE\_TEXT =  
 'Transfer failed: Insufficient balance';

```

        TransferFailed: insufficient balance';

ELSE

    UPDATE Account
    SET Balance = Balance - TransferAmount
    WHERE Account_No = SenderAccount;

    UPDATE Account
    SET Balance = Balance + TransferAmount
    WHERE Account_No = ReceiverAccount;

END IF;

END //

DELIMITER ;
CALL TransferMoney(10001, 10002, 5000);

```

Output: -

|   |    |          |                                          |                   |           |
|---|----|----------|------------------------------------------|-------------------|-----------|
| ✓ | 34 | 15:08:19 | CREATE PROCEDURE TransferMoney( IN Se... | 0 row(s) affected | 0.04      |
| ✓ | 35 | 15:08:53 | CALL TransferMoney(10001, 10002, 5000)   | 1 row(s) affected | 0.031 sec |

CHECK

```

SELECT *
FROM Account
WHERE Account_No IN (10001,10002);

```

Output: -

|   | Account_No | Customer_ID | Account_Type | Balance  | Branch     |
|---|------------|-------------|--------------|----------|------------|
| ▶ | 10001      | 101         | Savings      | 47000.00 | Hyderabad  |
|   | 10002      | 102         | Savings      | 80000.00 | Vijayawada |
| * | NULL       | NULL        | NULL         | NULL     | NULL       |

Query 18: - PROCEDURE WITH IN, OUT PARAMETERS

DELIMITER //

```
CREATE PROCEDURE GetBalance(  
    IN p_Account_No INT,  
    OUT p_Balance DECIMAL(12,2)  
)  
  
BEGIN  
  
    SELECT Balance  
    INTO p_Balance  
    FROM Account  
    WHERE Account_No = p_Account_No;  
  
END //
```

DELIMITER ;

```
CALL GetBalance(10001, @CurrentBalance);  
SELECT @CurrentBalance;
```

Output: -

|   |    |          |                                          |                   |
|---|----|----------|------------------------------------------|-------------------|
| ✓ | 37 | 15:11:44 | CREATE PROCEDURE GetBalance( IN p_Acc... | 0 row(s) affected |
| ✓ | 38 | 15:11:48 | CALL GetBalance(10001, @CurrentBalance)  | 1 row(s) affected |

| Result Grid |                 | Filter R |
|-------------|-----------------|----------|
|             | @CurrentBalance |          |
| ▶           | 47000.00        |          |

Query 19: -TRIGGER COMMANDS

```
SHOW TRIGGERS;
```

Output: -

| Result Grid              |        | Filter Rows      |                                      | Export:     |                        | Wrap Cell Content:  |     |  |  |
|--------------------------|--------|------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------------------------|--|
| Trigger                  | Event  | Table            | Statement                            | Timing                                                                                                                                                                                                                                                                                                                                                  | Created                | sql_mode                                                                                               | De  |                                                                                       |  |
| ▶ CheckTransactionAmount | INSERT | bank_transaction | BEGIN IF NEW.Amount <= 0 THEN SIG... | SIG... BEFORE                                                                                                                                                                                                                                                                                                                                           | 2026-08-15 14:59:57.20 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...                                                               | rod |                                                                                       |  |

Query 20: - PROCEDURE MANAGEMENT COMMANDS

```

SHOW PROCEDURE STATUS
WHERE Db = 'BankDB';
DROP PROCEDURE GetAllCustomers;

```

Output: -

|   | Db     | Name                | Type      | Definer        | Modified            | Created             | Security_type | Comment | character_set_client | collation_connection | Databa: Collation |
|---|--------|---------------------|-----------|----------------|---------------------|---------------------|---------------|---------|----------------------|----------------------|-------------------|
| ▶ | bankdb | DepositMoney        | PROCEDURE | root@localhost | 2026-08-15 14:55:53 | 2026-08-15 14:55:53 | DEFINER       |         | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4           |
|   | bankdb | GetBalance          | PROCEDURE | root@localhost | 2026-08-15 15:11:44 | 2026-08-15 15:11:44 | DEFINER       |         | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4           |
|   | bankdb | GetCustomerAccounts | PROCEDURE | root@localhost | 2026-08-15 14:52:47 | 2026-08-15 14:52:47 | DEFINER       |         | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4           |
|   | bankdb | TransferMoney       | PROCEDURE | root@localhost | 2026-08-15 15:08:19 | 2026-08-15 15:08:19 | DEFINER       |         | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4           |
|   | bankdb | WithdrawMoney       | PROCEDURE | root@localhost | 2026-08-15 14:57:56 | 2026-08-15 14:57:56 | DEFINER       |         | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4           |